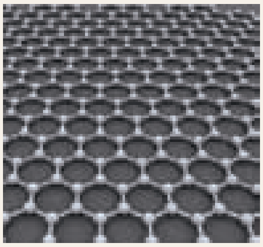
Super-fast communication using Graphene based optical modulator



Graphene is a single atom layer of Carbon atoms. Layers of Graphene are what ultimately gives us graphite. The fermi level of electrons in a sheet can be manipulated using voltage. While Graphene is naturally opaque to near-infrared light used in optical networks, it becomes transparent with a negative or positive voltage applied to it. Using this properly a team of researchers, led by UC Berkley Engineering professor Xiang Zhang, have developed an optical modulator with potential frequencies of up to 500 GHz, which is ten times faster than the current technologies. Now, an optical modulator is the major workhorse in an optical network, and the time in which it can switch between on and off is what generally decides the speed of operation of the network. An optical modulator of such high frequency of operation might just be perfect companion for Silicon photonics and our ultra-fast hard-drives. And since the internet also relies on an optical network, expect some glitch-free streaming of superHD content soon.

inside me has about 46 billion transistors – all of them three dimensional. Silicon was getting too hot with these many transistors – so I was outfitted with Graphene-based transistors instead. They are cool, literally.

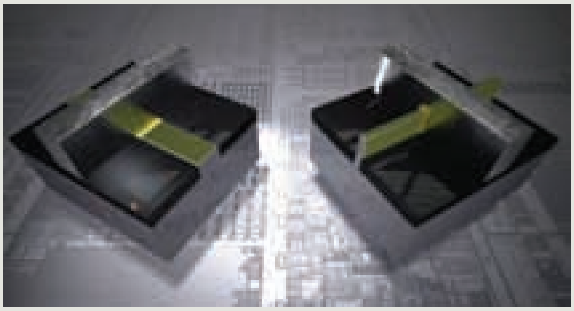
Everything besides the CPU is also closer to it. Like everything else, the PCs come in a neat package. Over the last decade, everything started entering this 'package', and now we have the CPU, GPU and RAM all bunched together like a cool family – literally again. Also, they don't communicate with each other through electrical signals. When the internals are so fast, the electrical signals were no longer feasible while switching, so they were replaced by the fastest signal known to man: Light. And the wonder substance of our generation, Graphene, is involved again. Which also means that it's been some five years since they stopped manufacturing the platter based hard drives, which were too slow. As spinning the platter any faster would have brought the world to an end, they were permanently replaced with solid state drives. Inside me, there is not even a port to plug-in those ancient hard drives. The Non-Volatile nature of the NVRAM also ensures that I boot-up immediately. What's the use of being super-fast if I take 100 seconds to get out of bed?

The communication with the outside components also happens with the same light based technology. You see, they weren't kidding when they said the future is bright #CoolPCJoke.

The peripherals outside the box are equally impressive – the display is a high-res holographic projection system which the GPU works hard to power. The make of the GPU? Well, Nvidia and AMD have still got their horns locked.

Unlike the rest of the world, the mouse and keyboard are still very much the same. Changing it in anyway would have led to a major decrease in the average WPM rates and a major economic decline, so the keyboard was left untouched for the general good of humanity. The mouse looks the same, but works on anything now – you can run it over water too. Even if you don't need to. d

3D Transistors



While everything is running off into the third dimension why should our beloved transistor be confined to a two dimensional existence. Intel recently announced the 3D transistor, which really is a fancy name for a tri-gate transistor. The raised fin (see figure) allows better control over current flow by having three contact surfaces with gate. This ensures that the current flow is high in 'ON' state and nearly zero in 'OFF' state. This better contact also ensures that the switching between these two states is fast. The good news is that this isn't a concept technology like most. This technology uses 22 nanometer manufacturing process, and will be used in 2012 processor range 'Ivy Bridge'. Look out for better efficiency, performance and low power requirements. Also expect Intel to try and wrestle the share in handheld market with this technology, where it has lost quite a lot of ground to ARM.